

NVIDIA JETSON THOR

GLOBANT PHYSICAL AI HACKATHON

Talk to your Company

A live 3D twin of your plants that you can talk to.

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Globant Physical AI Hackathon • October 2026

F3



F2



SECTION 1

The problem

01

Nobody can watch every camera



Events go unseen

Near-misses, restricted zones and missing helmets slip by.



Answers are slow

"What is happening on floor 3?" means walking there.



Walls are mute

Camera walls show video. They cannot be asked anything.



Cloud is a poor fit

Factory video off site is slow, costly and often not allowed.

The manager needs **one view of the whole site**, and **a way to ask it things**.

A live 3D twin of your company that you can talk to.

Ask by voice. The twin flies to the right floor, shows what the cameras see, and answers out loud.

SECTION 2

What it does

02



The whole company, live in one view.

Every person and vehicle the cameras see is a live marker on the right floor.

WHAT IT DOES

You ask in plain language, by voice or by text

YOU ASK

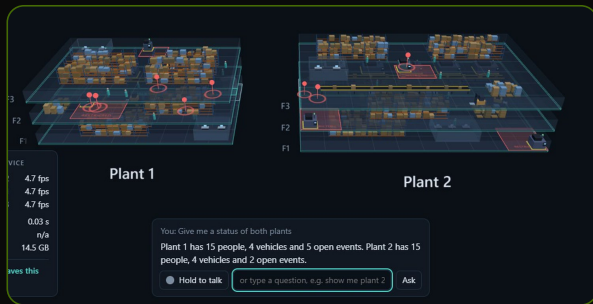
“Give me a status of both plants.”

YOU ASK

“Show me Plant 2, floor 3.”

YOU ASK

“Any safety issues in the last ten minutes?”



Answers are spoken, shown as text, and move the twin.

Four rules turn movement into safety events



Near miss

Person within 2 m of a closing forklift.



Restricted zone

Person inside for more than 2 seconds.



No helmet

The vision check on a person says no.



Crowding

More than 4 people within 3 m.

Events land on the **right floor**, are stored, and can be asked about by plant, floor, type and time.

WHAT IT DOES

Every floor has a live camera, with detections boxed



Focus a floor, see its camera

The view opens beside the twin.



One detection, two views

What places the marker also draws the box.



Nothing extra is exposed

One stream at a time, on the one app port.

Feed any camera from a recording or a webcam



Recorded video

mp4 files, looped as live streams.



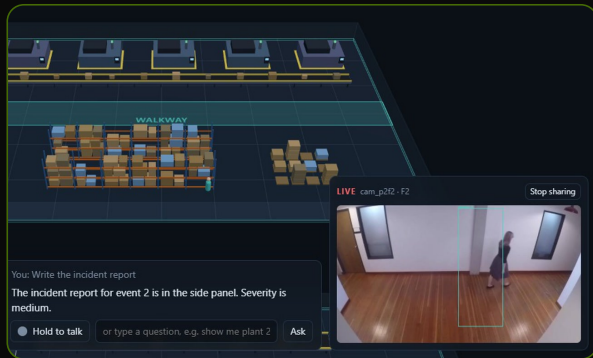
A webcam

Shared from the browser, or attached to the machine.



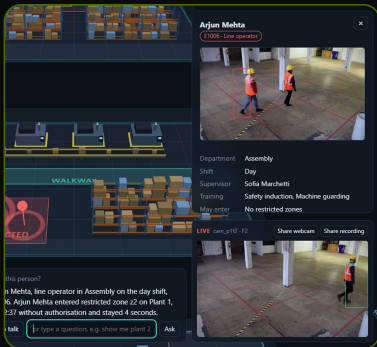
Share recording

Pick any video file; it is analysed on the spot.



WHAT IT DOES

The alert names the person. Ask who it is.



Who

Name, role, shift, supervisor and training.



When and how long

Entered, alert raised, left, time in the zone.



Allowed or not

Authorisation, and other incidents today.

SYNTHETIC EMPLOYEES

SIMULATED BADGE FEED

NO FACE RECOGNITION

One more sentence writes the incident report

YOU ASK

“Write the incident report.”



Complete

What, where, when, who, severity, action.



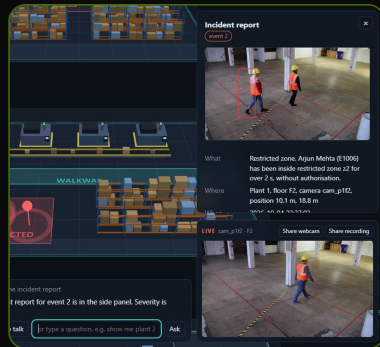
Grounded

Filled from the stored event, nothing invented.



Evidence attached

The camera snapshot from that moment.



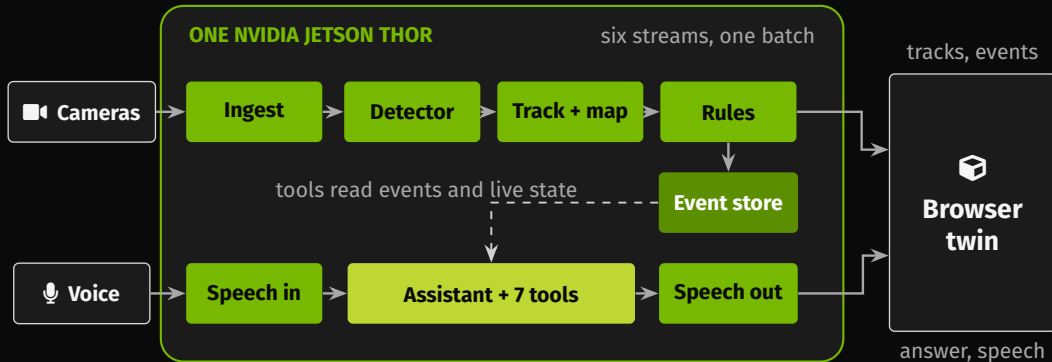
SECTION 3

How it works

03

HOW IT WORKS

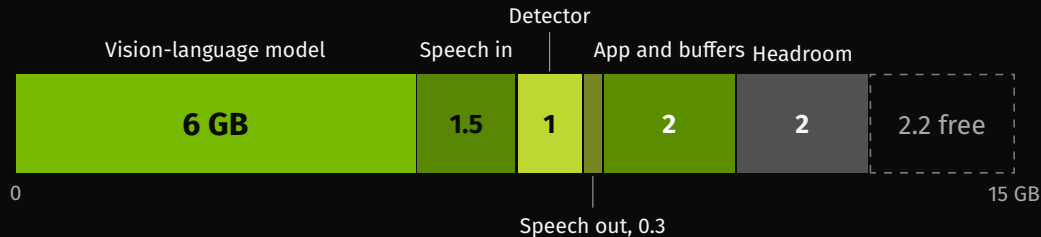
One device takes video to a spoken answer



One vision-language model serves the conversation and the “look at this frame” tool.

HOW IT WORKS

The planned model stack fits inside 15 GB



Vision model

3B, half precision.



Whisper and Piper

Speech stays local.



YOLO on TensorRT

One batch, six streams.

Planned budget. It still has to be measured on the Jetson.

Every heavy component has a mock



Perception

REAL

Video and
detector

MOCK

Simulated
people



Assistant

REAL

Vision model

MOCK

Keyword router



Speech in

REAL

Whisper

MOCK

A fixed question



Speech out

REAL

Piper

MOCK

A chime

A backend that fails to load falls back to its mock, so the demo cannot die.

SECTION 4

Status and demo

04

Proven on a laptop. The Jetson run is the next step.

77

automated tests passing

6

camera streams at once,
on a laptop CPU

0

runs on a Jetson so far

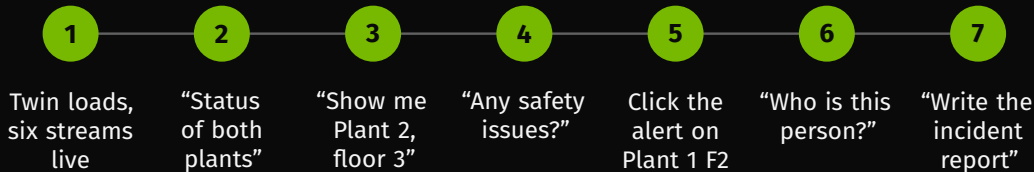
SHOWN WORKING

- ✓ Twin, events, conversation, report
- ✓ Real detection and speech on CPU
- ✓ Webcam and shared recording

STILL TO PROVE ON THE JETSON

- ⌚ Hardware decode and TensorRT
- ⌚ 3B model picks tools, 13 of 15
- ⌚ 10 fps on 6 streams, 2.5 s voice

The demo is seven steps



Then one unscripted question from a judge.

Why it matters



Safety

Events are seen, named and written up in the moment.



Operations

One person can ask about any floor without walking to it.



Privacy and cost

Audio and video stay on one device. No cloud calls.



Fit

A new site is a new configuration file, not a new project.

Ask your company a question. It shows you, and it tells you.

Talk-to-your-Company. Globant Physical AI Hackathon, built for one NVIDIA Jetson Thor.